Fake News Classifier

Kailash Shankar R [23BLC1008]

Qualitative Analysis

Background / Motivation

The rapid proliferation of digital media platforms has significantly amplified the spread of misinformation, particularly in the form of fake news. Detecting fake news has emerged as a critical challenge due to its far-reaching societal, political, and economic consequences. Traditional approaches based on manual fact-checking are increasingly inadequate, given the massive scale, speed, and volume at which information is generated and disseminated online. As a result, automated solutions leveraging Natural Language Processing (NLP) and Machine Learning (ML) techniques have become essential for scalable and efficient detection.

Furthermore, the complexity of fake news detection is heightened by the evolving and adversarial nature of misinformation. Modern fake content is often deliberately engineered to mimic the tone, structure, and credibility of legitimate journalism, making it difficult to distinguish using surface-level content analysis alone. Such articles may contain factually plausible statements while subtly manipulating context, emphasis, or narrative framing. This limitation highlights the need for hybrid approaches that analyze not only the semantic content of text but also its stylistic and structural characteristics. By incorporating linguistic, syntactic, and psychological indicators, automated systems can better uncover underlying patterns of deception and intent.

In this context, fake news differs from real news not only in factual accuracy but also in writing style, linguistic patterns, and emotional tone. This project aims to exploit these differences by combining content-based features, such as Term Frequency–Inverse Document Frequency (TF-IDF), with higher-level linguistic and structural features including part-of-speech distributions, sentence complexity, sentiment polarity, and readability metrics. By integrating multiple complementary feature representations, the proposed system captures both semantic meaning and stylistic nuances, thereby enabling a more robust and comprehensive approach to fake news classification.

Objectives

1. To develop a machine learning model for binary classification of news articles into fake and real categories
2. To extract and analyze both textual content features and linguistic stylistic features
3. To evaluate the effectiveness of combined feature representations over traditional TF-IDF alone
4. To visualize high-dimensional feature space using dimensionality reduction techniques such as t-SNE
5. To perform detailed error analysis to understand model limitations and misclassification patterns
6. To analyze the impact of feature scaling, normalization, and dimensionality reduction techniques on model performance and interpretability
7. To investigate the role of sentiment and readability features in distinguishing between emotionally driven fake news and structured real news articles
8. To evaluate model behavior using advanced visualization techniques such as correlation matrices and feature importance analysis to gain insights into decision-making patterns

Methods

1. Dataset and Data Preparation

The proposed system follows a comprehensive and structured Natural Language Processing (NLP) pipeline, beginning with data acquisition and culminating in model evaluation and analysis. The dataset utilized for this study is the ISOT Fake News Dataset, which contains labeled news articles categorized into two classes: real and fake. Each instance consists primarily of textual content along with associated metadata; however, for the purposes of this study, only the textual content and corresponding labels are retained to ensure a focused analysis on linguistic and semantic characteristics.

To eliminate potential biases arising from class imbalance, the dataset is balanced by randomly sampling an equal number of instances from both classes. This ensures that the model does not develop a bias toward the majority class and that evaluation metrics remain reliable and meaningful. Additionally, irrelevant or missing entries are removed to maintain data quality and consistency.

2. Text Preprocessing and Normalization

The preprocessing stage involves transforming raw textual data into a structured and analyzable format. Sentence tokenization is first performed using NLTK's pretrained Punkt tokenizer, which segments each document into individual sentences based on learned punctuation patterns. This is followed by word tokenization, where each sentence is further decomposed into individual tokens, including words and punctuation symbols.

Subsequently, lemmatization is applied using WordNet-based lemmatizers to reduce words to their canonical base forms. This step helps in reducing vocabulary size while preserving semantic meaning, thereby improving model generalization. Unlike conventional preprocessing pipelines that aggressively remove stopwords, this system employs a selective stopword strategy. Common function words are removed to reduce noise; however, pronouns are intentionally retained, as they serve as important indicators of writing style and author perspective, particularly in distinguishing persuasive or subjective content often associated with fake news.

3. Feature Engineering: Textual Representation

To enable machine learning algorithms to process textual data, each document is converted into a numerical representation. This is achieved using Term Frequency–Inverse Document Frequency (TF-IDF) vectorization. Unlike simple bag-of-words models, TF-IDF assigns weights to words based on their frequency within a document and their rarity across the corpus, thereby emphasizing informative and discriminative terms.

$$TF(t,d) = \frac{number\ of\ times\ t\ appears\ in\ d}{total\ number\ of\ terms\ in\ d}$$

$$IDF(t) = log\frac{N}{1+df}$$

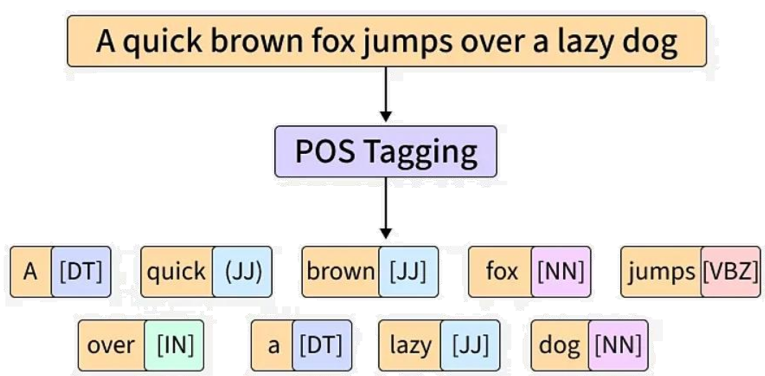
$$TF-IDF(t,d) = TF(t,d) * IDF(t)$$

Furthermore, the vectorizer incorporates n-gram modeling, specifically unigrams and bigrams, to capture contextual relationships between adjacent words. This allows the model to recognize meaningful phrases such as "breaking news" or "fake report," which may not be evident when considering individual words in isolation. The resulting representation is a high-dimensional sparse matrix, where each dimension corresponds to a unique term or phrase in the vocabulary.

4. Linguistic Feature Extraction

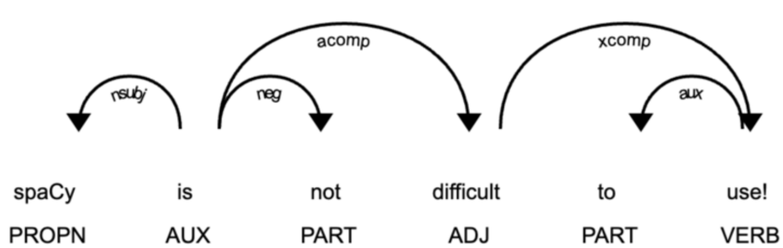
In addition to content-based features, the system extracts a range of linguistic features that capture stylistic and grammatical patterns. Part-of-Speech (POS) tagging is performed using NLTK's pretrained tagger, assigning grammatical categories such as nouns, verbs, adjectives, and adverbs to each token.

From these tags, several normalized features are computed. These include the superlative ratio, which measures the frequency of exaggerated expressions; the proper noun ratio, which reflects the specificity of named entities; the pronoun ratio, which captures the degree of subjectivity or persuasive tone; and adjective density, which indicates descriptive richness. These features are normalized by document length to ensure comparability across articles of varying sizes.



5. Structural and Syntactic Analysis

To further capture the complexity of language, structural features are incorporated. The average sentence length is calculated as the mean number of tokens per sentence, providing an estimate of readability and writing complexity. Additionally, syntactic tree depth is computed using spaCy's dependency parser. Each sentence is represented as a hierarchical tree structure, and the depth of this tree reflects the level of grammatical complexity. Real news articles are generally expected to exhibit deeper and more complex syntactic structures compared to simpler, more direct language often found in fake news.



6. Sentiment and Readability Features

To enrich the feature set, sentiment and readability metrics are included. Sentiment polarity is computed using TextBlob, which assigns a score ranging from -1 to +1, indicating negative to positive emotional tone. This helps capture the emotional intensity often associated with sensationalized or misleading content.

Readability is measured using the Flesch Reading Ease score via the textstat library. This metric quantifies how easy a text is to read based on sentence length and word complexity. Fake news articles are often designed to be easily digestible, whereas real news may exhibit more formal and complex language structures.

Flesch's Reading ease is calculated as:

$$206.835 - 1.015\left(\frac{\text{total words}}{\text{total sentences}}\right) - 84.6\left(\frac{\text{total syllables}}{\text{total words}}\right)$$

7. Feature Integration and Representation

All extracted features—TF-IDF vectors, linguistic metrics, structural attributes, sentiment scores, and readability measures—are combined into a unified feature representation. This is achieved using sparse matrix stacking, which efficiently merges high-dimensional sparse TF-IDF features with lower-dimensional dense linguistic features. The resulting feature matrix captures both semantic content and stylistic nuances, providing a comprehensive representation of each document.

8. Model Training and Evaluation Setup

The dataset is partitioned into training and testing subsets using an index-based splitting strategy. This approach ensures that both TF-IDF and combined feature matrices are split consistently, preventing data misalignment. The model selected for classification is Logistic Regression, chosen for its computational efficiency, robustness, and interpretability in high-dimensional spaces.

Logistic Regression is applied as follows:

$$P(y = 1|x) = \frac{1}{1+e^{-(\beta_0+\beta_1x_1+\beta_2x_2+\cdots+\beta_nx_n)}}$$

Where $P(y = 1|x)$ is the probability of the dependent variable being 1 given the independent variables $x_1, x_2, \ldots, x_n$.

Two models are trained: one using only TF-IDF features and another using the combined feature set. This allows for a comparative analysis of the contribution of linguistic and structural features to overall performance.

9. Performance Evaluation and Visualization

Model performance is evaluated using a comprehensive set of metrics, including accuracy, precision, recall, and F1-score. A confusion matrix is generated to analyze classification errors in detail, while the Receiver Operating Characteristic (ROC) curve is used to assess the model's ability to distinguish between classes across varying thresholds.

Metric	Formula
True positive rate, recall	$\frac{\text{TP}}{\text{TP+FN}}$
False positive rate	$\frac{\text{FP}}{\text{FP+TN}}$
Precision	$\frac{\text{TP}}{\text{TP+FP}}$
Accuracy	$\frac{\text{TP+TN}}{\text{TP+TN+FP+FN}}$
F-measure	$\frac{2 \cdot \text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}}$

Additionally, t-Distributed Stochastic Neighbor Embedding (t-SNE) is employed to visualize high-dimensional feature representations in a two-dimensional space, providing insights into clustering behavior and class separability. Finally, detailed error analysis is conducted by examining misclassified samples and comparing their feature profiles with class averages, enabling a deeper understanding of model limitations and decision boundaries.

Results

The proposed fake news classification system demonstrates strong overall performance across multiple evaluation metrics, indicating its effectiveness in distinguishing between fake and real news articles. The model achieves an overall accuracy of approximately 96–97%, reflecting a high level of correctness in predictions on unseen data. The classification report further reveals that both classes are handled effectively, with precision and recall values consistently exceeding 0.94. The F1-scores for both fake and real news remain balanced, indicating that the model does not exhibit significant bias toward either class and maintains stable performance across different evaluation measures.

- The confusion matrix illustrates that the majority of instances are correctly classified, with a high number of true positives and true negatives. Only a small number of misclassifications are observed, indicating that the model is capable of capturing discriminative patterns between the two classes. The Receiver Operating Characteristic (ROC) curve further supports these findings, with the curve positioned close to the top-left corner, demonstrating a strong trade-off between true positive rate and false positive rate and confirming the model's high discriminative capability.
- Visualization of the feature space using t-Distributed Stochastic Neighbor Embedding (t-SNE) provides additional insight into the separability of the data. The plot reveals the formation of distinguishable clusters corresponding to fake and real news articles, although some overlap is observed between the two classes. This overlap suggests that certain articles share similar linguistic and structural characteristics despite belonging to different categories.
- Feature-level analysis highlights the contribution of multiple linguistic and structural attributes in classification. The correlation heatmap indicates that no single feature dominates the prediction process, suggesting that the model relies on a combination of features rather than any individual metric. Sentiment and readability distributions further illustrate variation in textual characteristics between fake and real news, reflecting differences in emotional tone and complexity of writing.
- The inclusion of engineered features alongside TF-IDF representations results in improved model performance compared to using TF-IDF alone, demonstrating the effectiveness of hybrid feature integration. Additionally, qualitative examination of misclassified samples reveals that errors typically occur in cases

where the feature profile of a sample closely aligns with that of the opposite class.

Overall, the results confirm that combining textual, linguistic, and structural features enables robust classification performance, while also highlighting the inherent challenges posed by overlapping feature distributions in fake news detection.

Conclusion

This study presents a comprehensive approach to fake news detection by integrating both content-based and linguistic feature representations within a machine learning framework. The results demonstrate that combining TF-IDF features with stylistic, structural, and semantic attributes—such as part-of-speech distributions, sentence complexity, sentiment polarity, and readability—significantly enhances classification performance compared to relying solely on textual content. The model achieves high accuracy and balanced precision and recall across both classes, indicating its effectiveness in distinguishing between fake and real news articles.

Visualization techniques, including t-SNE and correlation analysis, reveal that while meaningful clustering exists in the feature space, there is a degree of overlap between classes, reflecting the inherent complexity of the problem. The inclusion of SHAP-based interpretability further strengthens the study by providing insights into how individual features contribute to model predictions, both at a global and instance-specific level. This highlights the importance of interpretability in machine learning applications, particularly in domains involving misinformation detection.

Despite strong performance, certain limitations remain. Misclassification occurs in cases where fake news closely mimics the linguistic and structural patterns of real news, indicating that style-based approaches alone may not fully capture factual correctness. Additionally, the reliance on manually engineered features may limit generalizability across diverse datasets and evolving misinformation patterns.

Future work can explore the incorporation of deep learning models, such as transformer-based architectures (e.g., BERT), to capture deeper semantic relationships. Integrating external knowledge sources and fact-checking mechanisms could further improve robustness. Overall, this work demonstrates the effectiveness of hybrid feature-based approaches while emphasizing the need for continued advancement in automated fake news detection systems.

Quantitative Analysis

2.1 Dataset Preprocessing

The dataset utilized in this study is the ISOT Fake News Dataset, which comprises a collection of labeled news articles categorized into two classes: fake (0) and real (1). Each instance consists of textual content representing news articles sourced from different domains. For the purpose of this study, only the textual component of each article is retained, as the focus is on analyzing linguistic and semantic characteristics.

To ensure balanced learning and avoid class bias, an equal number of samples (1000 instances each) are randomly selected from both fake and real news categories. The dataset is then shuffled to eliminate ordering bias and improve generalization.

Dataset Summary Table

Attribute	Description
Dataset Name	ISOT Fake News Dataset
Total Samples Used	2000
Class Distribution	1000 Fake, 1000 Real
Data Type	Text (News Articles)
Labels	Binary (0 = Fake, 1 = Real)

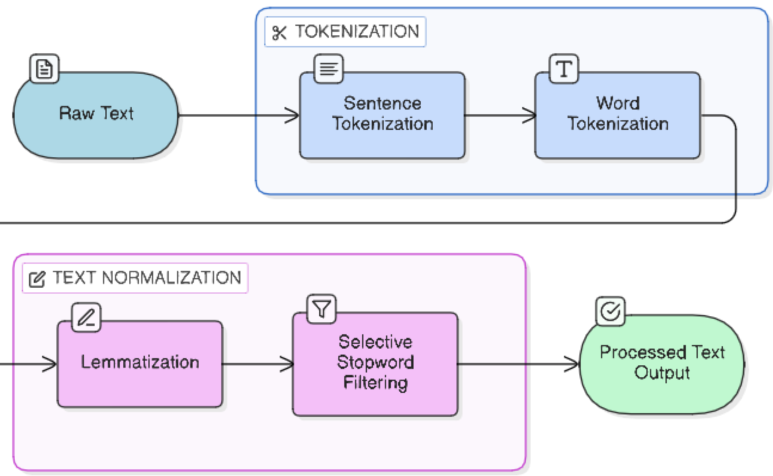
Preprocessing Pipeline

The preprocessing stage transforms raw text into structured input suitable for feature extraction. Sentence tokenization is first performed using the Punkt tokenizer from the Natural Language Toolkit (NLTK), which segments text into individual sentences. This is followed by word tokenization, where each sentence is further divided into tokens, including words and punctuation.

Subsequently, lemmatization is applied using WordNet-based lemmatizers, reducing each word to its base or dictionary form. This step reduces vocabulary size while preserving semantic meaning, thereby improving model efficiency.

Unlike conventional approaches that remove all stopwords, a selective stopword strategy is adopted. While common non-informative words are removed, pronouns are retained due to their importance in capturing stylistic and subjective elements of writing. This is particularly relevant for fake news detection, where persuasive language plays a key role.

Preprocessing Workflow



2.2 *Model Design / Algorithm Implementation*

The proposed system adopts a hybrid feature-based machine learning approach, combining both statistical text representations and linguistically informed features to improve classification performance.

Feature Representation

1. TF-IDF Vectorization

Textual data is transformed into numerical representation using Term Frequency–Inverse Document Frequency (TF-IDF). This method assigns weights to words based on their importance within a document and across the corpus.

To capture contextual relationships, n-gram modeling is incorporated, specifically unigrams and bigrams. This enables the model to recognize meaningful phrases rather than isolated words. The resulting representation is a high-dimensional sparse matrix.

2. Linguistic Features

In addition to TF-IDF, a set of linguistic features is extracted using Part-of-Speech (POS) tagging:

Superlative Ratio (JJS, RBS)
Proper Noun Ratio (NNP)
Pronoun Ratio (first-person vs third-person)
Adjective Count (JJ, JJR, JJS).

These features capture stylistic patterns and writing tendencies.

3. Structural Features
Average Sentence Length
Syntactic Tree Depth (using spaCy dependency parsing)
These features provide insight into grammatical complexity.

4. Additional Features
Sentiment Polarity (TextBlob)
Readability Score (Flesch Reading Ease)

Feature Integration
All features are combined into a single representation using sparse matrix stacking:

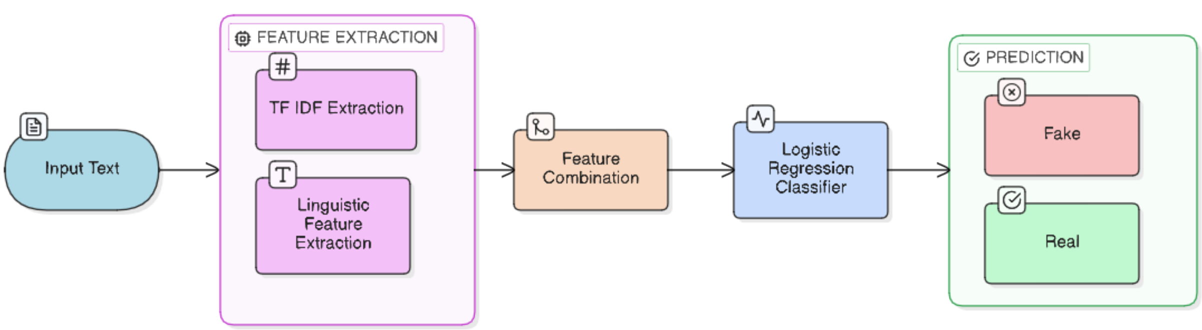
$$Combined\ Features = TF\text{-}IDF \oplus Linguistic \oplus Structural \oplus Sentiment \oplus Readability$$

Model Architecture
The classification model used is Logistic Regression, a linear probabilistic classifier suitable for high-dimensional data.

Key Hyperparameters

Parameter	Value
Model	Logistic Regression
Max Iterations	1000
TF-IDF Max Features	~7000
N-gram Range	(1,2)

Model Workflow



2.3 Training and Evaluation Setup

The dataset is split into training and testing subsets using an 80:20 ratio. Instead of directly splitting feature matrices, index-based splitting is employed to ensure consistency across multiple feature representations.

Data Split

Dataset Split	Percentage
Training Set	80%
Testing Set	20%

Training Process

The model is trained on:

- TF-IDF features (baseline model)
- Combined features (enhanced model)

This enables comparative evaluation of feature effectiveness.

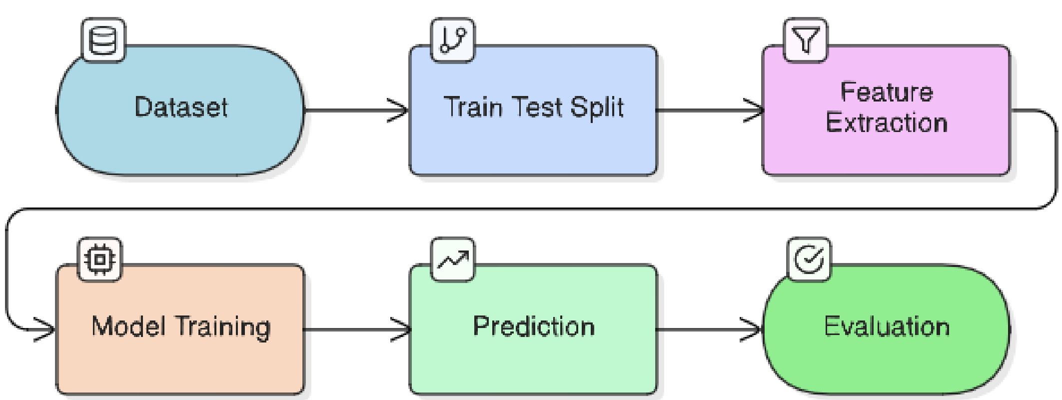
Evaluation Metrics

The model is evaluated using the following metrics:

Metric Formula

Metric	**Formula**
Accuracy	(TP + TN) / (TP + TN + FP + FN)
Precision	TP / (TP + FP)
Recall	TP / (TP + FN)
F1-score	2 × (Precision × Recall) / (Precision + Recall)

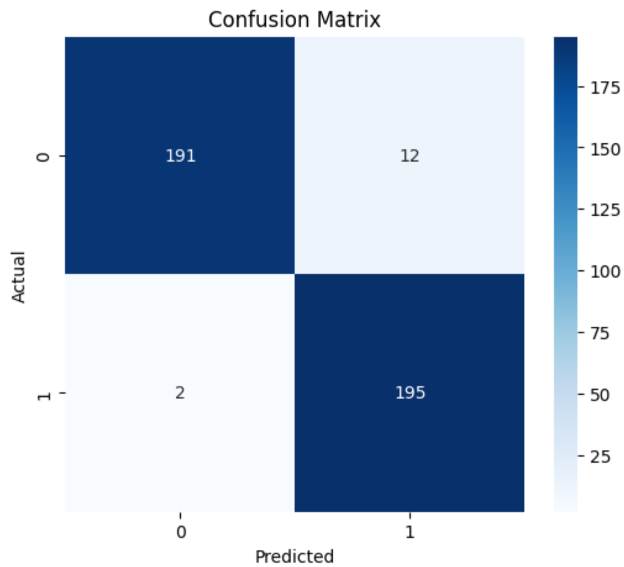
Training Pipeline



2.4 Experimental Results

1. Confusion Matrix

Given figure presents the confusion matrix of the proposed classification model. The matrix indicates that the model correctly classifies the majority of instances in both classes, with a high number of true positives and true negatives. Only a small number of misclassifications are observed, demonstrating strong predictive performance across both fake and real news categories.



2. Classification Report

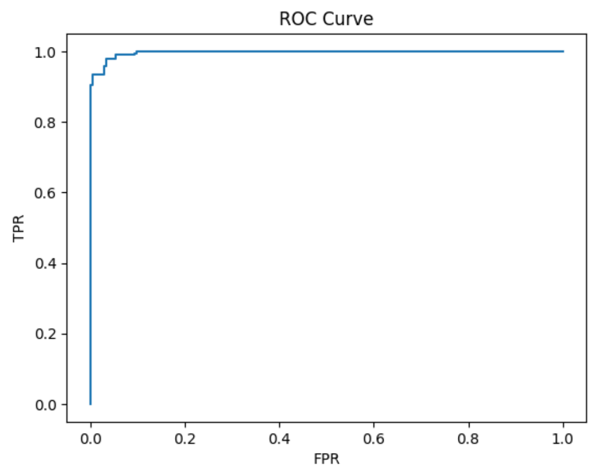
The classification report summarizes the model's performance in terms of precision, recall, and F1-score for both classes. The model achieves high precision and recall values for both fake and real news, with an overall accuracy of approximately 96%. The results indicate balanced performance across classes with minimal variation in F1-scores.

Classification Report:

	precision	*recall*	*f1-score*	*support*
0	*0.99*	*0.94*	*0.96*	*203*
1	*0.94*	*0.99*	*0.97*	*197*
accuracy			*0.96*	*400*
macro avg	*0.97*	*0.97*	*0.96*	*400*
weighted avg	*0.97*	*0.96*	*0.96*	*400*

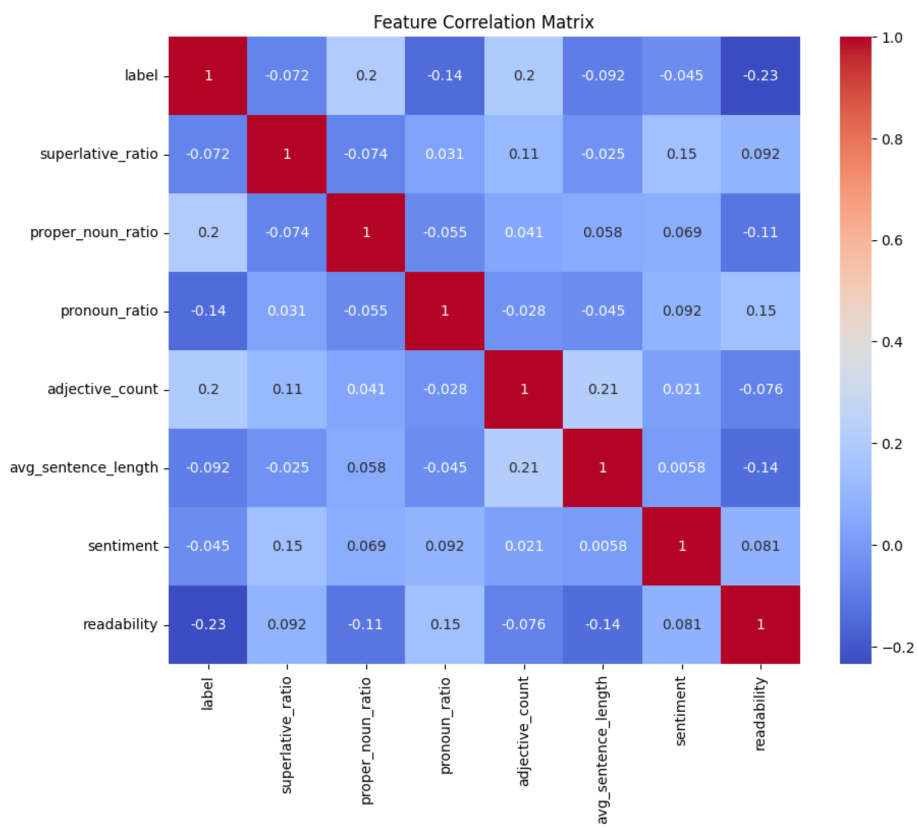
3. Roc Curve

Given figure illustrates the Receiver Operating Characteristic (ROC) curve for the classification model. The curve is positioned close to the top-left corner, indicating a high true positive rate and low false positive rate across different threshold values. This reflects strong discriminative capability of the model.



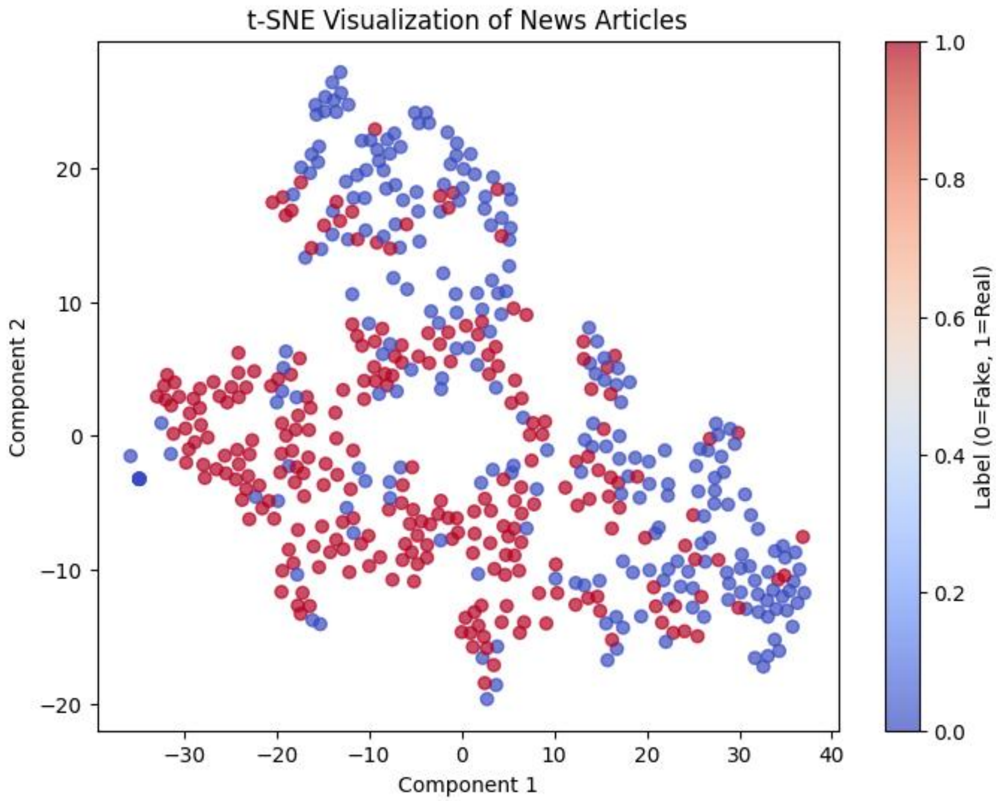
4. Correlation Heatmap

Given figure presents the correlation matrix of the extracted features. The heatmap indicates varying degrees of correlation between features and the target label. No single feature exhibits a strong correlation with the label, suggesting that classification relies on a combination of multiple features.



2.5 Visual Results

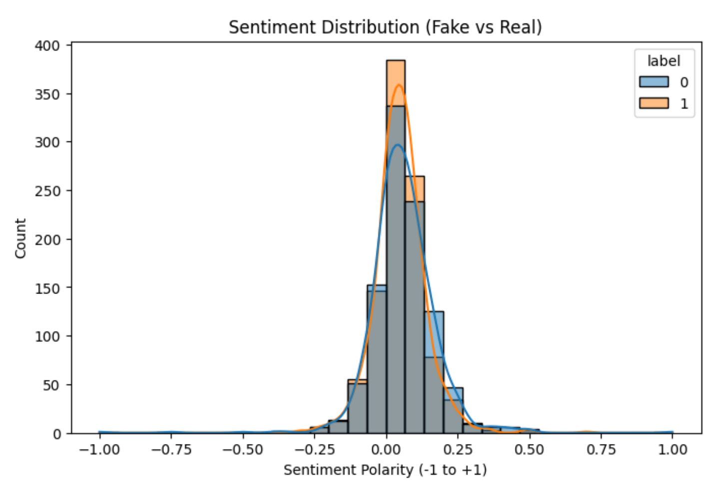
1. t-SNE Visualization



Given figure shows the t-Distributed Stochastic Neighbor Embedding (t-SNE) visualization of the dataset in two-dimensional space. The plot reveals distinct clusters corresponding to fake and real news articles, with some regions exhibiting overlap between the two classes.

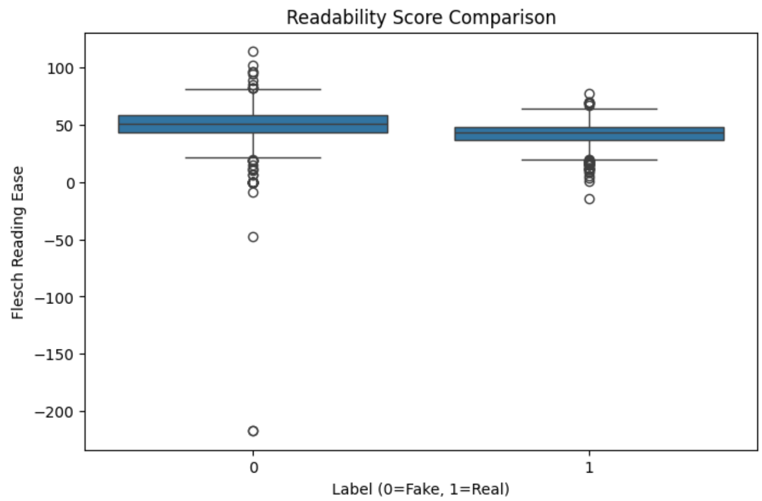
2. Sentiment Distribution

Given figure depicts the distribution of sentiment polarity for fake and real news articles. The plot shows variation in sentiment values across both classes, with differences in distribution patterns observable between fake and real news.



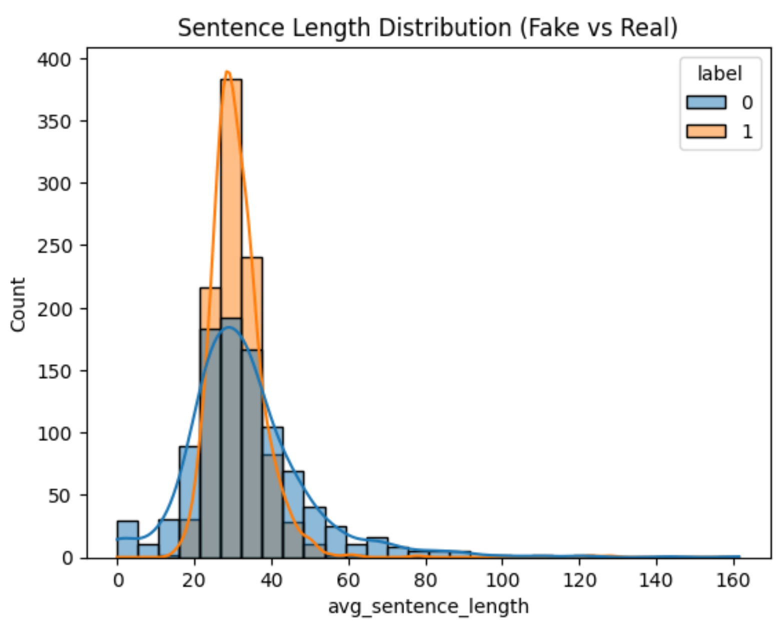
3. Readability Boxplot

Given figure illustrates the distribution of readability scores for fake and real news articles. The boxplot highlights differences in central tendency and spread between the two classes, indicating variation in text complexity.



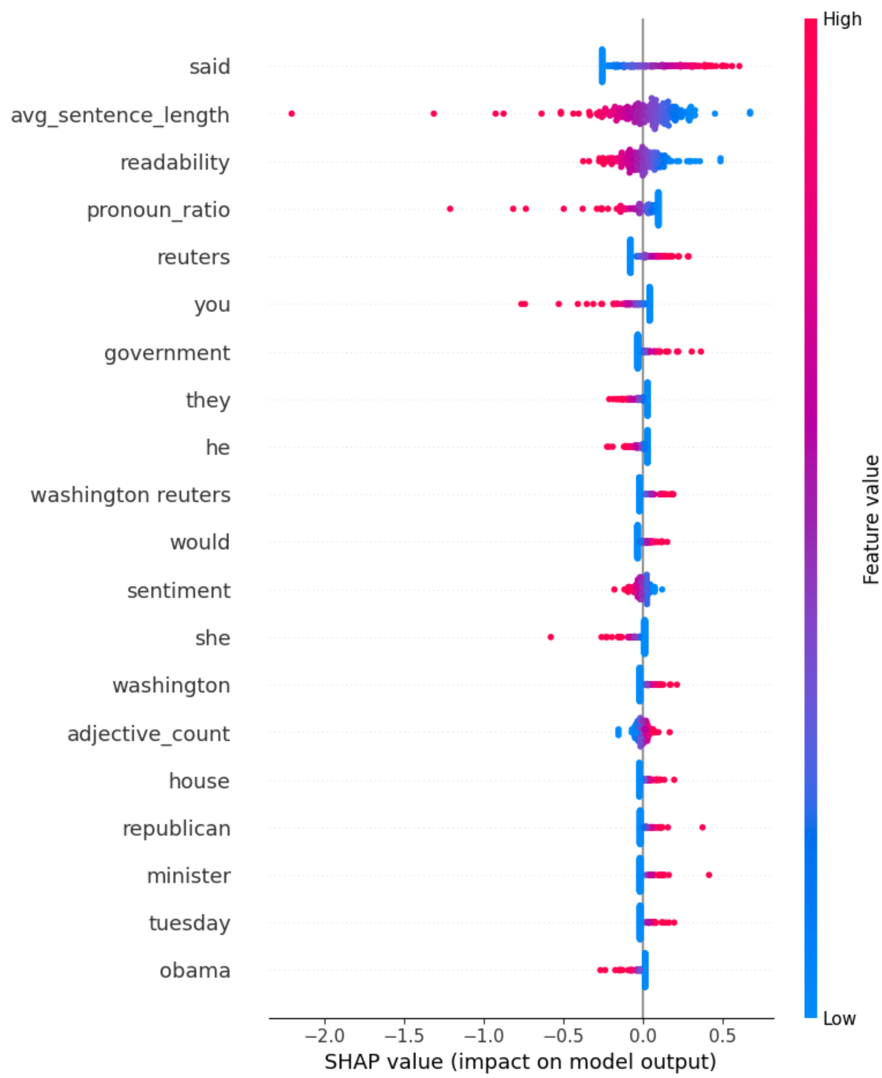
4. Sentence Length Distribution

Given figure shows the distribution of average sentence length across fake and real news articles. The histogram indicates variability in sentence length across both classes, with observable differences in distribution patterns.



5. Error Analysis And SHAP Graph

The presented figure illustrates the normalized feature values of a randomly selected misclassified sample in comparison with the average feature values of fake and real news classes. The visualization highlights the relative positioning of the sample across multiple linguistic and structural features, enabling a direct comparison with class-level distributions. In addition to this, SHAP (SHapley Additive Explanations) visualization is employed to analyze the contribution of individual features to the model's prediction for the selected sample. The SHAP plot provides a detailed representation of how specific features influence the classification outcome by indicating their positive or negative impact on the predicted class. Together, these visualizations offer a comprehensive view of both the statistical feature alignment and the model-driven feature contributions for misclassified instances.



Analysis and Discussions

The experimental outcomes collectively suggest that the proposed model does not merely memorize surface-level lexical patterns but learns a nuanced interplay between content, style, and structure in distinguishing fake and real news. The high overall performance, as reflected in the evaluation metrics, should not be interpreted solely as an indication of model strength, but rather as evidence that the chosen feature space successfully captures discriminative characteristics inherent to the problem domain. In particular, the balance between precision and recall across both classes indicates that the model is not disproportionately biased toward either fake or real news, which is critical in applications where both false positives and false negatives carry significant consequences.

- A deeper examination of the **confusion matrix** reveals that misclassifications are not random but systematically concentrated in specific regions of the feature space. These errors typically arise when fake news articles exhibit linguistic patterns that closely resemble those of legitimate journalism. This observation aligns with the evolving nature of misinformation, where deceptive content is intentionally crafted to mimic the tone and credibility of authentic news sources. Consequently, the model's occasional inability to distinguish such instances highlights a fundamental challenge in fake news detection: the boundary between classes is not strictly defined by content accuracy alone, but by subtle stylistic cues that may overlap.
- **The ROC curve** further reinforces the model's strong discriminative capability across varying thresholds, suggesting that the learned decision boundary is robust and adaptable. However, when interpreted alongside the t-SNE visualization, a more complex picture emerges.
- While the ROC curve indicates effective separability in a probabilistic sense, the **t-SNE plot** reveals that the underlying feature space contains regions of overlap between fake and real news clusters. This apparent contradiction underscores an important insight: separability in high-dimensional space does not always translate to clear visual separation in reduced dimensions. Instead, it suggests that the model relies on subtle, multidimensional relationships between features that are not easily captured in two-dimensional projections.
- **The correlation heatmap** provides further evidence of the distributed nature of feature importance. The absence of strong individual correlations with the target variable indicates that no single feature is sufficient to drive classification

decisions. Instead, the model leverages a combination of features, each contributing incrementally to the final prediction. This is particularly significant in the context of fake news detection, where reliance on a single indicator could lead to brittle models that fail under adversarial conditions. The distributed contribution of features enhances robustness, as it reduces the likelihood of the model being misled by isolated patterns.

- **Sentiment and readability analyses** offer additional layers of insight into the stylistic differences between fake and real news. While the distributions of sentiment polarity overlap, subtle shifts in their spread suggest that fake news may exhibit more extreme or varied emotional tones. Similarly, differences in readability scores indicate variation in linguistic complexity, with fake news often adopting simpler or more accessible language structures. However, these differences are not absolute, which explains why sentiment and readability alone are insufficient for classification. Instead, they function as complementary signals that enhance the overall feature representation.
- The **analysis of sentence length** further supports the notion that structural characteristics play a role in distinguishing classes. Variations in sentence length distributions suggest differences in writing style and complexity, with real news potentially exhibiting more structured and formal sentence constructions. Nevertheless, the overlap in distributions indicates that structural features alone cannot fully separate the classes, reinforcing the importance of integrating multiple feature types.
- One of the most critical insights emerges from the **error analysis of misclassified samples**. By comparing the feature profile of a misclassified instance with class-level averages, it becomes evident that such samples often occupy intermediate positions in the feature space. These instances do not strongly align with the typical characteristics of either class, making them inherently ambiguous. This observation highlights a key limitation of supervised classification: the model can only learn patterns present in the training data, and ambiguous or borderline cases will inevitably challenge its decision boundaries.
- The incorporation of **SHAP-based interpretability** provides a deeper understanding of the model's decision-making process. Unlike traditional feature importance measures, SHAP values offer a localized explanation of how individual features contribute to specific predictions. The SHAP summary plot reveals that both lexical features (e.g., specific words) and engineered linguistic features (e.g., sentence length, readability) play significant roles in influencing

predictions. This confirms that the hybrid feature approach is effective in capturing both semantic and stylistic dimensions of the data.

- More importantly, **SHAP analysis demonstrates that feature contributions are context-dependent.** The same feature may have different impacts depending on its value and interaction with other features. For example, a high value of a particular feature may push the prediction toward one class in some instances, while contributing differently in others. This context-sensitive behavior reflects the complexity of natural language and underscores the limitations of simplistic, rule-based approaches.

From a broader perspective, the results illustrate that fake news detection is inherently a **multi-dimensional problem** that cannot be addressed through a single type of feature or model. The success of the proposed approach lies in its ability to integrate diverse feature representations, enabling the model to capture subtle patterns that span lexical, syntactic, and semantic dimensions. At the same time, the presence of overlapping feature distributions and misclassified samples highlights the limitations of purely text-based approaches, suggesting the need for incorporating external knowledge or contextual information in future work.

In conclusion, the analysis demonstrates that while the model achieves strong performance, its behaviour is shaped by the complex interplay of multiple features and the inherent ambiguity of the data. The findings emphasize the importance of hybrid feature engineering, robust evaluation, and interpretability in developing reliable fake news detection systems.

References

1. Horne, B. D., & Adalı, S. (2017). *This just in: Fake news packs a lot in title, uses simpler, repetitive content in text body, more similar to satire than real news*. Proceedings of the International AAAI Conference on Web and Social Media.
2. Shu, K., Sliva, A., Wang, S., Tang, J., & Liu, H. (2017). *Fake news detection on social media: A data mining perspective*. ACM SIGKDD Explorations Newsletter, 19(1), 22–36.
3. Lazer, D. M. J., Baum, M. A., Benkler, Y., et al. (2018). *The science of fake news*. Science, 359(6380), 1094–1096.
4. Pennington, J., Socher, R., & Manning, C. D. (2014). *GloVe: Global vectors for word representation*. Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing (EMNLP).
5. Ramos, J. (2003). *Using TF-IDF to determine word relevance in document queries*. Proceedings of the First Instructional Conference on Machine Learning.
6. Pedregosa, F., Varoquaux, G., Gramfort, A., et al. (2011). *Scikit-learn: Machine learning in Python*. Journal of Machine Learning Research, 12, 2825–2830.
7. Bird, S., Klein, E., & Loper, E. (2009). *Natural Language Processing with Python*. O'Reilly Media.
8. Lundberg, S. M., & Lee, S.-I. (2017). *A unified approach to interpreting model predictions*. Advances in Neural Information Processing Systems (NeurIPS).
9. Mikolov, T., Chen, K., Corrado, G., & Dean, J. (2013). *Efficient estimation of word representations in vector space*. arXiv preprint arXiv:1301.3781.
10. Van der Maaten, L., & Hinton, G. (2008). *Visualizing data using t-SNE*. Journal of Machine Learning Research, 9, 2579–2605.
11. Devlin, J., Chang, M.-W., Lee, K., & Toutanova, K. (2019). *BERT: Pre-training of deep bidirectional transformers for language understanding*. NAACL.